# Global C-MAPSS RUL Study

Train on every FD001–FD004 training trajectory; evaluate the locked model on every official FD001–FD004 test set

Prepared 20 October 2026 | CPU-bounded local execution | 709 training engines | 707 official test engines

## Protocol

The four training files were concatenated after remapping engine identifiers. Three HistGradientBoosting candidates were selected using only combined-training data: three subset-stratified engine-grouped folds and fixed 10/25/50/75/100/125-cycle pseudo-censored endpoints. The condition-normalized candidates fit six operating-condition clusters and sensor moments on development rows in each fold. Official test labels were read only after the validation choice was fixed.

## Global validation selection

| Candidate | Cap | Condition norm. | RMSE | MAE | R2 | PHM08 mean |
|---|---|---|---|---|---|---|
| global_hgb_cap100_condition_normalized | 100 | yes | 14.101 | 9.730 | 0.878 | 2.532 |
| global_hgb_cap125_condition_normalized | 125 | yes | 11.800 | 8.178 | 0.914 | 2.561 |
| global_hgb_cap125_raw_sensor_control | 125 | no | 17.169 | 12.526 | 0.819 | 7.546 |

Selected before official testing: **global_hgb_cap100_condition_normalized**. It is ranked by mean PHM08, followed by RMSE. The cap and normalizer are then refit on all 709 training engines before endpoint prediction.

## Locked official-test results

| Test subset | Engines | RMSE | MAE | R2 | PHM08 mean | Bias |
|---|---|---|---|---|---|---|
| FD001 | 100 | 16.292 | 11.608 | 0.846 | 3.164 | -7.688 |
| FD002 | 259 | 31.128 | 19.183 | 0.665 | 61.477 | -14.767 |
| FD003 | 100 | 17.273 | 11.982 | 0.826 | 3.785 | -6.686 |
| FD004 | 248 | 33.391 | 21.785 | 0.625 | 62.574 | -17.451 |

## Pooled result

Across all 707 official test engines: RMSE **28.737**, MAE **18.006**, R2 **0.683**, and PHM08 mean **45.454**. This aggregate is descriptive; the per-subset rows remain the primary result because fault-mode and operating-condition coverage differs across FD001–FD004.

## Interpretation and next step

This is the requested all-training-data model, evaluated consistently on every official test file. It measures performance after exposure to both single- and multi-condition data and both one- and two-fault training regimes. It should be compared with the earlier FD004-only transfer study as a separate experiment, because that study intentionally withheld FD001–FD003 training data. The saved endpoint predictions support further error analysis, calibration review, or a lightweight edge-model distillation experiment.

## Reproducibility

Runner: run_all_subsets_global_study.py. The complete validation comparison, per-subset endpoint predictions, global model artifact, split inventories, and machine-readable metrics are stored in reports/all_subsets_global and artifacts/all_subsets_global.